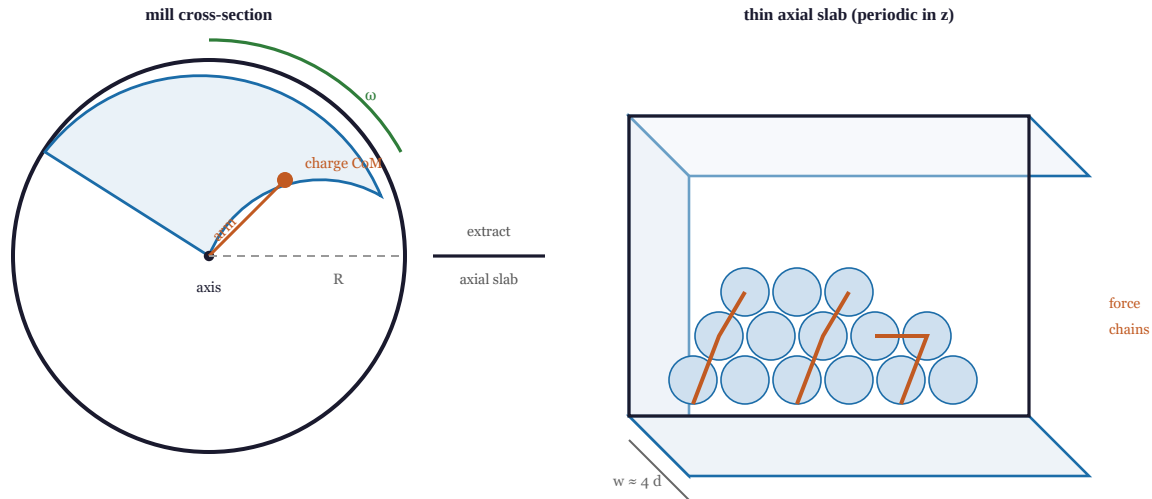


The thin-3D slab: why the torque arm scales with mill radius



2D disc: lift is a size-independent absolute height, so arm / R shrinks with size. 3D slab: force chains carry the drive, so arm scales with R (dynamic similarity).

Net power (van Nierop torque route): $P = 2\pi T N$, scaled by length / w .